MA1521 L1 Week 09 Mon 20261012

Evaluate $\displaystyle\iint_R \frac{x-3y}{2x+y}\,dx\,dy$

R is the parallelogram bounded on the sides by
$y = -2x+1$ and $y = -2x+4$,
and above and below by
$y = x/3$ and $y = (x-7)/3$.

Use the change of variables $u = x - 3y$, $v = 2x + y$.

Evaluate $\iint_R \cos\left(\frac{x-y}{x+y}\right)\,dx\,dy$

R is the triangle with vertices at the origin, (2,0) and (1,1).

Evaluate $\iint_R (2x - 3y)^2(x + y)^2 \, dx \, dy$

R is the triangle bounded by the positive x-axis, negative y-axis, and line $2x - 3y = 4$.

Use the change of variables $u = x + y$, $v = 2x - 3y$.

Evaluate $\iint_R (x^2 + y^2)\, dx\, dy$

R is the finite region in the first quadrant bounded by the lines $y = x$ and $y = 2x$, and a portion of the hyperbola $xy = 3$.